Swayam Mehta

Raleigh, NC | (919)-885-8701 | swayamehta1@gmail.com | linkedin.com/in/swayam-mehta

PROFESSIONAL SUMMARY

Forward Deployed Engineer–oriented Machine Learning and Software Engineer with experience deploying AI-powered systems from prototype through production and engaging directly with customers through demos, pilots, and technical walkthroughs. Comfortable translating complex technical systems into clear business value for diverse audiences. Strong background in backend systems, AI/ML integration, DevOps pipelines, and production validation, with a bias for ownership, iteration, and real-world impact.

SKILLS

Programming & Scripting: Python, C/C++, Java, JavaScript, Bash
AI & Data Science: Machine learning, deep learning, computer vision, predictive modeling, feature engineering
AI Frameworks: PyTorch, TensorFlow (basic), OpenCV, NumPy, Pandas
Cloud & Deployment: AWS, Azure, Azure DevOps, Docker, Linux
Data & Analytics: SQL, PostgreSQL, MongoDB, large dataset processing
Engineering Tools Exposure: AutoCAD, thermal/optical cameras, sensor systems, RTSP pipelines
Visualization & BI: Dashboards, data visualization, reporting, Power BI (basic)
Collaboration: Agile methods, backlog prioritization, cross-functional teamwork

CERTIFICATION

- **Complete Terraform and Ansible Bootcamp - Udemy**

EXPERIENCE

Junior Systems Engineer — **May 2024 – Present**
SAM Analytic Solutions - SSAM Team — Raleigh, NC

- Worked directly with customers and internal stakeholders to deploy AI-driven monitoring and analytics systems into production environments.
- Planned and delivered live technical demonstrations, proof-of-value pilots, and solution walkthroughs for customers, tailoring depth and messaging to technical and non-technical audiences.
- Represented engineering at industry conferences and client events, manning booths, demonstrating AI platforms, and answering detailed technical questions from end users and decision-makers.
- Gathered real-time customer feedback during demos and deployments and fed insights back to product and engineering teams to influence feature improvements and roadmap priorities.
- Designed and maintained Python-based backend services and APIs supporting end-to-end AI workflows, from data ingestion to inference and operational reporting.
- Built evaluation and validation tooling to measure performance, robustness, and system behavior under real-world operating conditions, ensuring deployment success criteria were met.
- Implemented CI/CD pipelines using Azure DevOps to automate testing, validation, and deployment, accelerating iteration cycles and reducing deployment friction.
- Authored technical documentation, deployment guides, and operational playbooks used to scale customer onboarding and repeatable deployments.
- Supported post-deployment troubleshooting and customer enablement, helping resolve issues early and reduce escalation risk.

PROJECTS

Financial Insight Engine (RAG for Financial Documents) — **December - 2026**
UNC Charlotte

- Built an offline Retrieval-Augmented Generation (RAG) application to analyze complex financial documents (10-K/10-Q, earnings reports) and answer questions with page-level source citations.
- Implemented a microservices architecture using FastAPI (backend API), Streamlit (chat UI + PDF uploads), and ChromaDB (vector store), orchestrated with Docker Compose.
- Integrated local LLM inference via Ollama (Llama 3) to ensure documents never leave the host environment, supporting privacy-sensitive workflows.
- Developed PDF ingestion and indexing pipelines (chunking + embeddings) to enable accurate retrieval and citation-grounded responses.
- Added automated tests using pytest (unit + integration) and implemented basic performance metrics (analysis latency) for monitoring and regression checks.

EDUCATION

Bachelor of Science in Computer Engineering - Machine Learning Concentration — **August 2020 – December 2024**
Department of Electrical and Computer Engineering - University Of North Carolina at Charlotte

- Minor in Mathematics